FINANCIAL DATA ANALYSIS



Power BI | DAX |Analysis

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INTRODUCTION

CREDIT CARD CUSTOMER ANALYTICS DASHBOARD

This dashboard provides a comprehensive view of key credit card performance metrics and customer behavior. It helps in identifying high-value customers, monitoring risk factors, measuring satisfaction, and tracking delinquency and churn indicators.

The insights derived from this analysis empower data-driven decision making to improve customer experience, reduce risk, and drive sustainable growth.





DATA OVERVIEW

The dataset contains customer-level credit card information, including demographic details, credit limits, transaction history, account balances, and financial behavior. It includes fields such as Client Number, Card Category, Annual Income, Credit Limit, Total Transaction Amount, Total Transaction Count, Avg Utilization Ratio, Total Revolving Balance, Interest Earned, Delinquent Account Status, Personal Loan Status, and Week Start Date. This data is used to analyze customer spending patterns, credit utilization, financial risk, and overall portfolio performance through interactive Power BI dashboards and DAX-based KPIs.

Business Objectives

- Analyze customer transaction patterns and overall credit card usage.
- Measure financial performance using key DAX-based KPIs.
- Monitor credit utilization and identify high-risk customers.
- Track transaction growth using Week-over-Week (WoW) and Month-over-Month (MoM) analysis.
- Evaluate customer behavior through churn, delinquency, and satisfaction metrics.
- Identify top-performing clients based on transaction amount.
- Support data-driven decision-making to improve customer retention, reduce financial risk, and optimize banking operations.

1. Running Total of Credit Card Transactions

</> DAX Measure

```
Running_total=
      CALCULATE
            [Total_Trans_Amt],
       FILTER
           ALL('calendar'),
           'calendar'[Date]<=MAX('calendar'[Date]) ))
```

</> Explanation

- ALL('calendar') - Removes any existing filters on the calendar table to consider all dates.
- 'calendar'[Date] <= MAX('calendar'[Date]) - Includes all dates up to the current context date.
- CALCULATE([Total_Trans_Amt], ...) - Recalculates the total transaction amount for the filtered date range.
- Result - Provides the cumulative (running) total of transaction amounts up to each date.

Running Total of Credit Card Transactions

Year	Month	Total Trans_Amt	Running_total
2023	January	4322186	4322186
2023	February	3539575	7861761
2023	March	3388827	11250588
2023	April	4174728	15425316
2023	May	3426913	18852229
2023	June	3533660	22385889
2023	July	4546958	26932847
2023	August	3449868	30382715
2023	September	3452874	33835589
2023	October	4050909	37886498
2023	November	3405420	41291918
Total		**45533021**	**45533021**



Insight: The running total shows a consistent upward trend in transaction amounts throughout the year 2023, indicating steady business growth.

2. Calculate the 4-week moving average of the creditLimit for each client.

</> DAX Measure

```
moving_average_4_week =
VAR time_period =
            FILTER(
                  ALL('calendar')  calendar' [week_num] >
      MAX('calendar' [week_num]) - 4   && 'calendar'[week_num] <=
       MAX('calendar'[week_num] )
 VAR credit_limt = CALCULATE([credit_limit], time_period)
 VAR distinct_count = CALCULATE(DISTINCTCOUNT('calendar' [week_num]),
                        time_period)
 RETURN
     DIVIDE(credit_limt, distinct_count, 0)
```

</> Explaination

- time_period - Creates a 4-week window including the current week and the previous 3 weeks.
- credit_limt - Calculates total credit limit for the selected 4-week period.
- distinct_count - Counts distinct weeks in the 4-week window.
- DIVIDE - Divides total credit limit by number of weeks to get the average, with 0
- as alternate result.

4-week moving average of the creditLimit for each client.

Year	week_num	credit_limit	moving Average_4 week
2023	1	17,04,635.70	17,04,635.70
2023	2	16,00,959.70	16,52,797.70
2023	3	15,26,415.10	16,10,670.17
2023	4	16,47,543.20	16,19,888.43
2023	5	17,98,645.00	16,43,390.75
2023	6	17,10,629.70	16,70,808.25
2023	7	19,76,267.00	17,83,271.23
2023	8	17,33,164.80	18,04,676.63
2023	9	15,07,517.70	17,31,894.80
2023	10	14,41,620.20	16,64,642.43
2023	11	16,48,313.70	15,82,654.10
Total		**8,89,56,376.00**	**16,28,887.03**

INSIGHT The 4-week moving average helps in smoothing out week-to-week volatility in credit limits,providing a clearer view of the underlying trend for better credit risk assessment and planning.

3. Calculate the mom% growth

</> DAX Measure

```
MOM_Growth =
 var prev_momth =
        CALCULATE([Total_Trans_Amt] ,
         DATEADD('calendar'[Date],-1,MONTH))

RETURN
 DIVIDE([Total_Trans_Amt]-prev_momth,prev_momth,0)
```

</> Explaination

- **DATEADD('calendar'[Date], -1, MONTH) - gets the transaction amount from the previous month.**
- **DIVIDE function calculates the percentage change from the previous month.**
- **Result shows the Month-over-Month (MoM) % growth.**

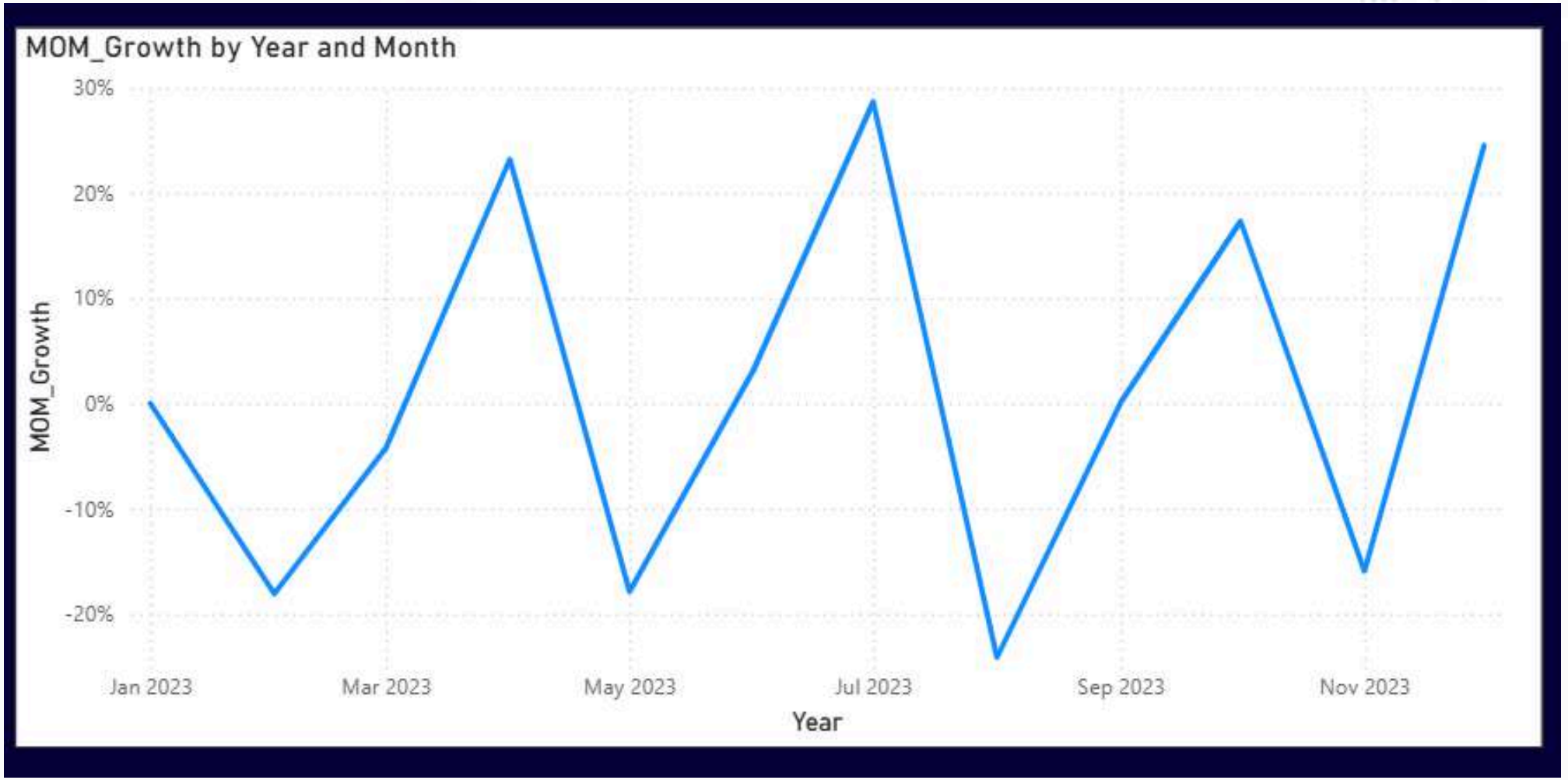
mom% growth

Year	Month	Total_Trans_Amt	MOM_Growth
2023	January	4322186	0.00
2023	February	3539575	-0.18
2023	March	3388827	-0.04
2023	April	4174728	0.23
2023	May	3426913	-0.18
2023	June	3533660	0.03
2023	July	4546958	0.29
2023	August	3449868	-0.24
2023	September	3452874	0.00
2023	October	4050909	0.17
2023	November	3405420	-0.16
2023	December	4241103	0.25
Total		**45533021**	**0.10**



INSIGHT : MoM growth analysis helps in understanding short-term trends,detecting fluctuations early, and supporting strategic financial planning.

3. Line Chart of mom% growth



INSIGHT : MoM growth shows extreme monthly volatility with peak growth in July 2023 (29%) and the steepest drop in August 2023 (24%).

3. Calculate the wow% growth

</> DAX Measure

```
WOW_Growth =
 var prev_week =
    CALCULATE([Total_Trans_Amt],
      FILTER(ALL('calendar'[week_num]),
      'calendar'[week_num] = MAX('calendar'[week_num])-1))

RETURN
  DIVIDE([Total_Trans_Amt]-prev_week,prev_week,0)
```

</> Explaination

- **FILTER with ALL removes existing filters and gets the transaction amount of the previous week.**
- **DIVIDE function calculates the percentage change from the**
- **previous week.**
- **Result shows the Week-over-Week (WoW) % growth.**

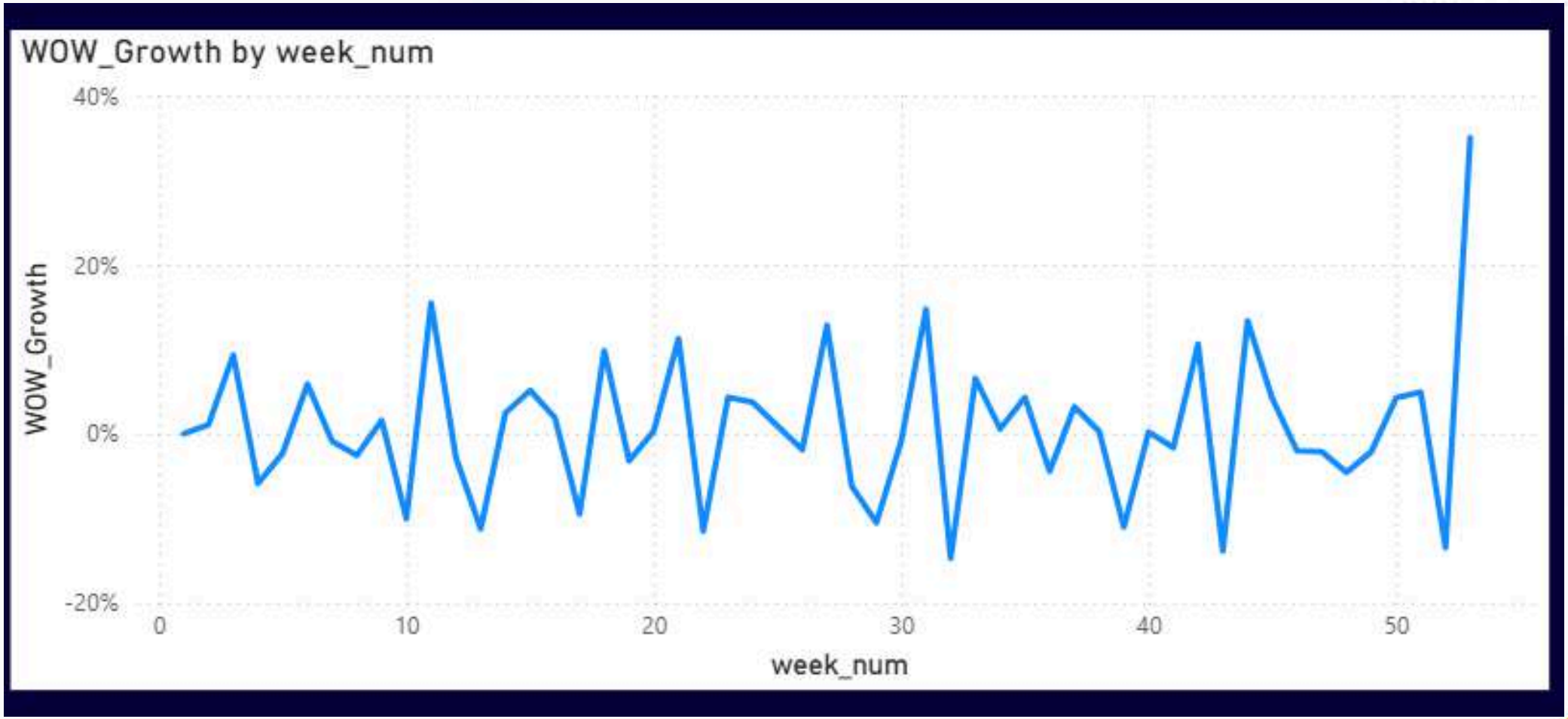
wow% growth

Year	week_num	Total_Trans_Amt	WOW_Growth
2023	1	835767	0.00%
2023	2	844739	1.07%
2023	3	923367	9.31%
2023	4	869235	-5.86%
2023	5	849078	-2.32%
2023	6	898867	5.86%
2023	7	890756	-0.90%
2023	8	868091	-2.54%
2023	9	881861	1.59%
2023	10	793080	-10.07%
2023	11	915725	15.46%
2023	12	890081	-2.80%
Total		**45533021**	**5981.80%**



INSIGHT : WoW growth analysis helps in understanding short-term trends,detecting fluctuations early, and supporting strategic financial planning.

3. Line chart of wow% growth



INSIGHT : WoW growth remains mostly stable between -10% and +15%, culminating in a major year-end spike of nearly 35% in Week 53..

4. Calculate Customer Acquisition Cost (CAC) as a Ratio of Transaction Amount.

Customer Acquisition Cost (CAC) as a Ratio of Transaction Amount.

Client_Num	customer_acq_cost	Total_Trans_Amt	CAC
708082083	87	15149	0.57%
708083283	108	992	10.89%
708084558	106	1447	7.33%
708085458	150	3940	3.81%
708086958	106	4369	2.43%
708095133	94	1448	6.49%
708098133	75	1598	4.69%
708099183	75	2732	2.75%
708100533	64	4943	1.29%
708103608	80	15798	0.51%
708104658	118	2928	4.03%
708108333	86	1507	5.71%
708112008	86	1661	5.18%
Total	**991160**	**45533021**	**2.18%**

</> DAX Measure

```
customer_acq_cost=
        SUM('credit card'[Customer_Acq_Cost])

CAC =
    DIVIDE([customer_acq_cost],[Total_Trans_Amt],0)
```

</> Explanation

- **customer_acq_cost - Calculates the total customer acquisition cost by summing the Customer_Acq_Cost column.**
- **CAC - Divides the total customer acquisition cost by the total transaction amount.**
- **Result - Represents the CAC as a percentage of the transaction amount, indicating acquisition cost efficiency.**

INSIGHT : A lower CAC ratio suggests more efficient customer acquisition, while a higher ratio indicates higher marketing spend relative to revenue.

5. Calculate the yearly average of avg_utilization_ratio for all clients.

</> DAX Measure

```
yearly_avg_util_ratio=
        CALCULATE(AVERAGE('creditcard'
            [Avg_Utilization_Ratio]),
          ALLEXCEPT('calendar','calendar'[year]))
```

</> Explaination

- **AVERAGE - Calculates the average of Avg_Utilization_Ratio across the filtered data.**
- **ALLEXCEPT('calendar', 'calendar'[year]) - Removes all filters from the calendar table except the Year filter, ensuring the average is calculated for each year.**
- **Result - Returns the yearly average of Avg_Utilization_Ratio for each client.**

yearly average of avg_utilization_ratio for all clients.

Client_Num	Year	yearly_avg_util_ratio
708082083	2023	46.90%
708083283	2023	73.60%
708084558	2023	21.40%
708085458	2023	0.00%
708086958	2023	23.90%
708095133	2023	5.50%
708098133	2023	50.00%
708099183	2023	32.70%
708100533	2023	85.00%
708103608	2023	21.20%
708104658	2023	61.90%
708108333	2023	0.00%
708112008	2023	4.50%
Total		**27.45%**



INSIGHT : This metric helps in identifying clients with higher or lower credit utilizationpatterns over the year for better risk assessment and planning.

6. Calculate the percentage of Interest_Earned compared to Total_Revolving_Bal for each client.

</> DAX Measure

```
perct_earned =
        DIVIDE(
            SUM('credit card'[Interest_Earned]),
        SUM('credit card'[Total_Revolving_Bal]))
```

</> Explaination

- **SUM('credit card'[Interest_Earned]) - Calculates the total interest Earned for each client.**
- **SUM('credit card'[Total_Revolving_Bal]) - Calculates the total Revolving Balance for each client.**
- **DIVIDE - Divides Interest Earned by Total Revolving Balance to get the percentage. Returns 0 if denominator is zero.**

Interest_Earned compared to Total_Revolving_Bal for each client.

Client_Num	Interest_Earned	Total_Revolving_Bal	perct_earned
708082083	4,393.21	1661	264.49%
708083283	69.44	2517	2.76%
708084558	202.58	1771	11.44%
708086958	1,004.87	749	134.16%
708095133	275.12	1833	15.01%
708098133	159.80	1418	11.27%
708099183	409.80	1873	21.88%
708100533	988.60	2277	43.42%
708103608	3,791.52	2517	150.64%
708104658	732.00	890	82.25%
708112008	465.08	1049	44.34%
708119658	554.18	1034	53.60%
708129933	884.10	1544	57.26%
708132783	785.80	1179	66.65%
Total			**66.63%**



INSIGHT : A higher percentage indicates stronger interest income relative to revolving balance,while a lower percentage may suggest low utilization or lower interest generation.

7. Calculate Top 5 Clients by Total Transaction Amount.

Top 5 Clients by Total Transaction Amount.

Client_Num	Total_Trans_Amt	Top_cust_by_trans
920819113	79463	1
919695363	19739	2
956622169	19597	3
941614504	18504	4
718140783	18484	5
717642633	17995	6
801036033	17744	7
716004258	17634	8
713758758	17628	9
712503408	17498	10
778428108	17437	11
Total	**45533021**	**1**

</> DAX Measure

```
Top_cust_by_trans =
        RANKX(ALL('credit card'),
            [Total_Trans_Amt],,DESC,Dense)
```

</> Explanation

- **RANKX -** Ranks clients based on [Total_Trans_Amt].
- **ALL('credit card') -** Removes all filters to rank across the entire dataset.
- **DESC -** Ranks in descending order (highest to lowest).
- **DENSE -** Ensures no gaps in ranking (1, 2, 3, 4, 5 ...).



INSIGHT : The top clients contribute significantly to the total transaction amount.Focusing on these high-value clients can drive more revenue and strengthen relationships.

8. Identify clients whose Avg_Utilization_Ratio exceeds 80%.

Avg_Utilization_Ratio exceeds 80%.

Client_Num	avg_utitlzation_ratio	avg_utilization_greater_80%
708082083	46.90%	No
708083283	73.60%	No
708084558	21.40%	No
708085458	0.00%	No
708086958	23.90%	No
708095133	5.50%	No
708098133	50.00%	No
708099183	32.70%	No
708100533	85.00%	Yes
708103608	21.20%	No
708104658	61.90%	No
708108333	0.00%	No
708112008	4.50%	No
Total	**27.45%**	**No**

</> DAX Measure

```
avg_utilization_greater_80% =
        IF(
            [avg_utitlzation_ratio]>0.80,"Yes","No"
    )
```

</> Explanation

- **[avg_utilization_rratio] > 0.80 - Checks if the client's average utilization ratio is greater than 80%.**
- **IF function - Returns "Yes" if the condition is true, otherwise returns "No".**
- **Result - Identifies high utilization clients who may be at higher credit risk.**

INSIGHT : Clients with Avg_Utilization_Ratio exceeding 80% may be over-reliant on credit. Monitoring these clients helps in proactive risk management and credit control.

9. Create a KPI that flags clients who have not made any transactions (Total_Trans_Amt = 0) in the last 6 months.

</> DAX Measure

```
churn_check =
var lpd = MAX('credit card'[Week_Start_Date])
var maxtd = CALCULATE
   (MAX('credit card'[Week_Start_Date]),ALL('credit card'))
return
  if(lpd>maxtd-180,1,0)
```

</> Explaination

- **lpd (Last Purchase Date) - Retrieves the most recent transaction date for the client.**
- **maxtd (Max Transaction Date) - Retrieves the maximum transaction date in the entire dataset.**
- **lpd > maxtd - 180 - Checks if the client's last transaction was more than 180 days (6 months) ago.**
- **Result (churn_check) - Returns 1 if the client is inactive for 6 months, otherwise returns 0.**

Customer Churn Indicator

Client_Num	lpd	maxTd	churn_check
708502908	01-01-2023	31-12-2023	0
708504783	01-01-2023	31-12-2023	0
708506508	01-01-2023	31-12-2023	0
708507483	08-01-2023	31-12-2023	0
708508758	08-01-2023	31-12-2023	0
708510858	08-01-2023	31-12-2023	0
708511983	08-01-2023	31-12-2023	0
708513183	08-01-2023	31-12-2023	0
708514383	08-01-2023	31-12-2023	0
708515358	08-01-2023	31-12-2023	0
708516558	08-01-2023	31-12-2023	0
708519108	08-01-2023	31-12-2023	0
708522033	08-01-2023	31-12-2023	0
708525108	08-01-2023	31-12-2023	0
708528483	08-01-2023	31-12-2023	0
708532308	08-01-2023	31-12-2023	0
708534033	08-01-2023	31-12-2023	0
708535158	08-01-2023	31-12-2023	0
708536283	08-01-2023	31-12-2023	0
708538608	08-01-2023	31-12-2023	0
708539583	08-01-2023	31-12-2023	0
Total	**31-12-2023**	**31-12-2023**	**1**



INSIGHT : Clients flagged with 1 (Inactive) have not transacted in the last 6 months. These clients may be at risk of churn and can be targeted with re-engagement campaigns.

10. Calculate the percentage of clients with Delinquent_Acc > 0.

</> DAX Measure

```
Delinquency_Rate =
                DIVIDE(
                    CALCULATE(COUNTROWS('credit card'),
                 'credit card'[Delinquent_Acc]>0),
                COUNTROWS('credit card'),0)
```

percentage of clients with Delinquent_Acc > 0.

6.06%

delinquent_rate

</> Explaination

- **CALCULATE(COUNTROWS('credit card'), 'credit card' [Delinquent_Acc] > 0) - Counts the number of clients who have one or more delinquent accounts (Delinquent_Acc > 0).**
- **COUNTROWS('credit card') - Counts the total number of clients in the dataset.**
- **DIVIDE(numerator, denominator, 0) - Divides the number of delinquent clients by the total number of clients to get the percentage. Returns 0 if divisor is 0.**



INSIGHT : A higher delinquency rate indicates higher credit risk. Monitoring this metric helps in early risk detection and better portfolio management.

11. Create a score for each client based on their Avg_Utilization_Ratio, Delinquent_Acc, and Total_Revolving_Bal.

</> DAX Measure

```
credit_card_risk =
var avg_utlization = IF(MAX('credit card'
                        [Avg_Utilization_Ratio])>0.80,40,
          IF(MAX('credit card'[Avg_Utilization_Ratio])>0.50,20,10))
var delinquent = IF(MAX('credit card'[Delinquent_Acc])>0,40,0)
var revolving_score = IF(MAX('credit card'
                         [Total_Revolving_Bal])>=2000,40,
          IF(MAX('credit card'[Total_Revolving_Bal])>1000,20,10))
return
   avg_utlization+delinquent+revolving_score
```

</> DAX Measure

```
risk_category =
          IF([credit_card_risk]>=80,"high risk",
            IF([credit_card_risk]>=50,"medium risk","low risk"))
```

Credit Risk Score:

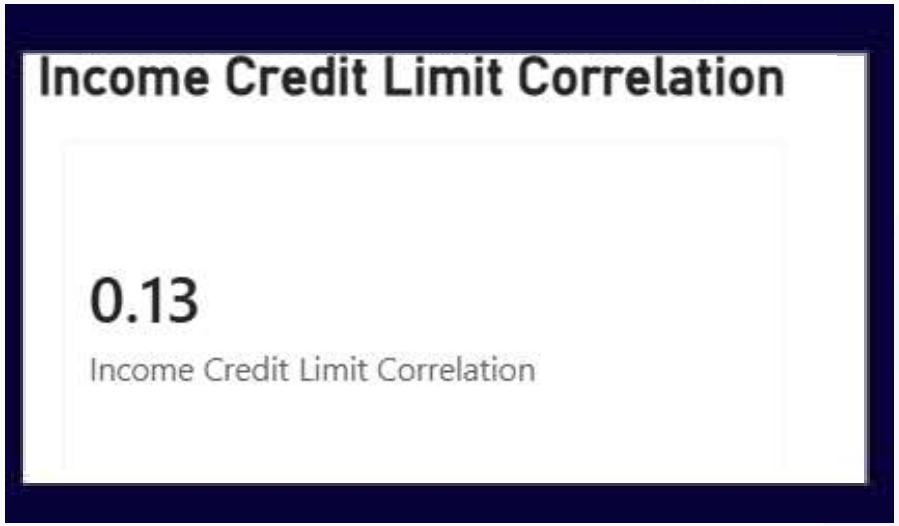
Client_Num	avg_utitlzation_ratio	Delinquent_Acc	Total_Revolving_Bal	credit_card_risk	risk_category
708083283	73.60%	0	2517	60	medium risk
708103608	21.20%	0	2517	50	medium risk
708147108	11.90%	0	2517	50	medium risk
708158133	83.00%	1	2517	120	high risk
708373083	75.00%	1	2517	100	high risk
708421833	75.30%	0	2517	60	medium risk
708445608	76.20%	1	2517	100	high risk
708474333	20.50%	0	2517	50	medium risk
708532308	22.00%	0	2517	50	medium risk
708536283	86.70%	0	2517	80	high risk
708555633	70.90%	0	2517	60	medium risk
708560733	67.80%	0	2517	60	medium risk
708580908	17.30%	0	2517	50	medium risk
708584208	80.90%	0	2517	80	high risk
Total	**27.45%**			**120**	**high risk**

INSIGHT : Clients with higher scores are at greater credit risk.
This helps in prioritizing monitoring, risk mitigation and targeted intervention strategies.

12. Income vs Credit Limit Correlation: Show the correlation between Income and Credit_Limit for all clients.

</> DAX Measure

```
Income Credit Limit Correlation =
    VAR ClientData = FILTER( ADDCOLUMNS
                        ( VALUES(Customer[Client_Num]), "Income",
          CALCULATE(MAX(Customer[Income])),
                        "CreditLimit",
          CALCULATE(MAX(Credit_Card[Credit_Limit])) ),
                        NOT ISBLANK([Income]) &&
          NOT ISBLANK([CreditLimit]) )
  VAR AvgIncome
       = AVERAGEX(ClientData, [Income])
  VAR AvgCreditLimit = AVERAGEX(ClientData, [CreditLimit])
  VAR Numerator = SUMX( ClientData, ([Income] - AvgIncome)
            * ([CreditLimit] - AvgCreditLimit) )
   VAR Denominator = SQRT( SUMX(ClientData, ([Income] -
                                    AvgIncome) ^ 2)
                * SUMX(ClientData, ([CreditLimit] - AvgCreditLimit) ^ 2) )
  RETURN DIVIDE(Numerator, Denominator)
```



Credit limit is nearly the same for clients with and without personal loans, suggesting that personal loan status alone may not strongly influence credit limits.

13. Calculate the average Cust_Satisfaction_Score by Card_Category.

</> DAX Measure

```
Avg_cust_satisfaction=
                    AVERAGE(customer
                            [Cust_Satisfaction_Score]
                    )
```

</> Explanation

- **AVERAGE() - calculates the mean of Cust_Satisfaction_Score.**
- **'customer'[Cust_Satisfaction_Score] - refers to the satisfactionscore column in the customer table.**
- **The measure is evaluated for each Card_Category in the visual,providing the average satisfaction score.**

average Cust_Satisfaction_Score by Card_Category.

Card_Category	cust_satisfaction_Average
Blue	3.20
Gold	3.05
Platinum	2.72
Silver	3.22
Total	**3.19**



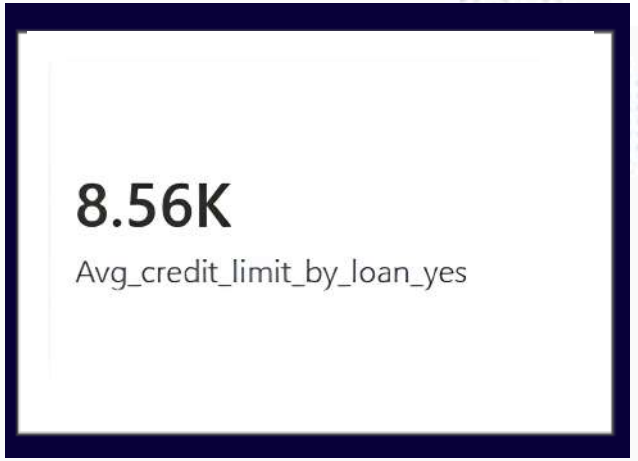
INSIGHT : Silver cardholders show the highest average satisfaction (3.22),followed closely by Blue (3.20). Platinum cardholders have the lowest average satisfaction (2.72).

14. Loan Approval vs Credit Limit:

</> DAX Measure

```
Avg_credit_limit_by_loan_yes =
                CALCULATE(
                 AVERAGE('credit card'[Credit_Limit]),
                  customer[Personal_loan] = "yes")

Avg_Credit_Limit_by_loan_No =
                CALCULATE(
                  AVERAGE('credit card'[Credit_Limit]),
                   customer[Personal_loan] = "no")
```





</> Explaination

- **The CALCULATE (): function filters clients based on their Personal Loan status (Yes or No), while AVERAGE() calculates the average Credit Limit for each group.**



Clients without personal loans have a slightly higher average credit limit (8,654.79) than those with personal loans (8,557.56), indicating only a small difference between the two groups.

15. Create a flag for clients whose Total_Revolving_Bal exceeds 90% of their Credit_Limit and who have a high Avg_Utilization_Ratio.

</> DAX Measure

```
High_Risk_client =
                    IF('credit card'[Total_Revolving_Bal] >
            0.9 * 'credit card'[Credit_Limit] &&
           'credit card'[Avg_Utilization_Ratio] > 0.80,
      "High Risk",
  "Low Risk")
```

</> Explanation

- Checks if Total_Revolving_Bal is greater than 90% of Credit_Limit.
- Checks if Avg_Utilization_Ratio is greater than 80%.
- Flags the client as "High Risk" if both conditions are true; otherwise "Low Risk".

High_Risk_Client

Client_Num	Total_Revolving_Bal	90%_of_credit	Credit_Limit	Avg_Utilization_Ratio	High_Risk_client
708083283	2517	3078.90	3,421.00	73.60%	Low Risk
708103608	2517	10708.20	11,898.00	21.20%	Low Risk
708147108	2517	18975.60	21,084.00	11.90%	Low Risk
708158133	2517	2727.90	3,031.00	83.00%	High Risk
708373083	2517	3020.40	3,356.00	75.00%	Low Risk
708421833	2517	3007.80	3,342.00	75.30%	Low Risk
708445608	2517	2974.50	3,305.00	76.20%	Low Risk
708474333	2517	11064.60	12,294.00	20.50%	Low Risk
708532308	2517	10277.10	11,419.00	22.00%	Low Risk
708536283	2517	2613.60	2,904.00	86.70%	High Risk
708555633	2517	3193.20	3,548.00	70.90%	Low Risk
708560733	2517	3342.60	3,714.00	67.80%	Low Risk
708580908	2517	13095.90	14,551.00	17.30%	Low Risk
708584208	2517	2799.90	3,111.00	80.90%	High Risk
708668283	2517	4626.00	5,140.00	49.00%	Low Risk
708672708	2517	31064.40	34,516.00	7.30%	Low Risk
708714108	2517	12096.90	13,441.00	18.70%	Low Risk
708809283	2517	9324.00	10,360.00	24.30%	Low Risk
708818508	2517	3107.70	3,453.00	72.90%	Low Risk
708822783	2517	2675.70	2,973.00	84.70%	High Risk
708833133	2517	2853.00	3,170.00	79.40%	Low Risk
708841233	2517	13213.80	14,682.00	17.10%	Low Risk
708890958	2517	28472.40	31,636.00	8.00%	Low Risk
Total		**80060738.40**			

INSIGHT : Clients flagged as High Risk are using more than 90% of their credit limit and have high utilization. These clients may require closer monitoringand proactive risk management.

THANK YOU !

Thank for your time and attention.
I hope this analysis provides valuable insights and support data-driven decision-making for Financial Analysis.